MicroBlaze-V-Zephyr-Kria-OpenAMP

Getting Started Guide, document revision v1.0

A. Preparing the KR260 and Sensor Setup

1. Purchase the KR260 and Pmod Accessories
    a. KR260 Board , power adapter , 16+ GB of SD Card – Purchase Link from AMD
    b. Pmod Acl – Digilent Link
    c. Pmod TMP2 – Digilent Link
    d. Pmod GPIO 8 bit LED – Digilent Link
    e. Pmod UART – Digilent Link
    f. Few Male to female jumpers – Amazon or can be purchased from any electronics shop.

2. Prepare the KR260 with power adapter , 16GB+ of SD card and burn the Kria-Ubuntu 22.04 LTS OS image on the SD card.
    a. Download image
    b. Burn using Balena etcher

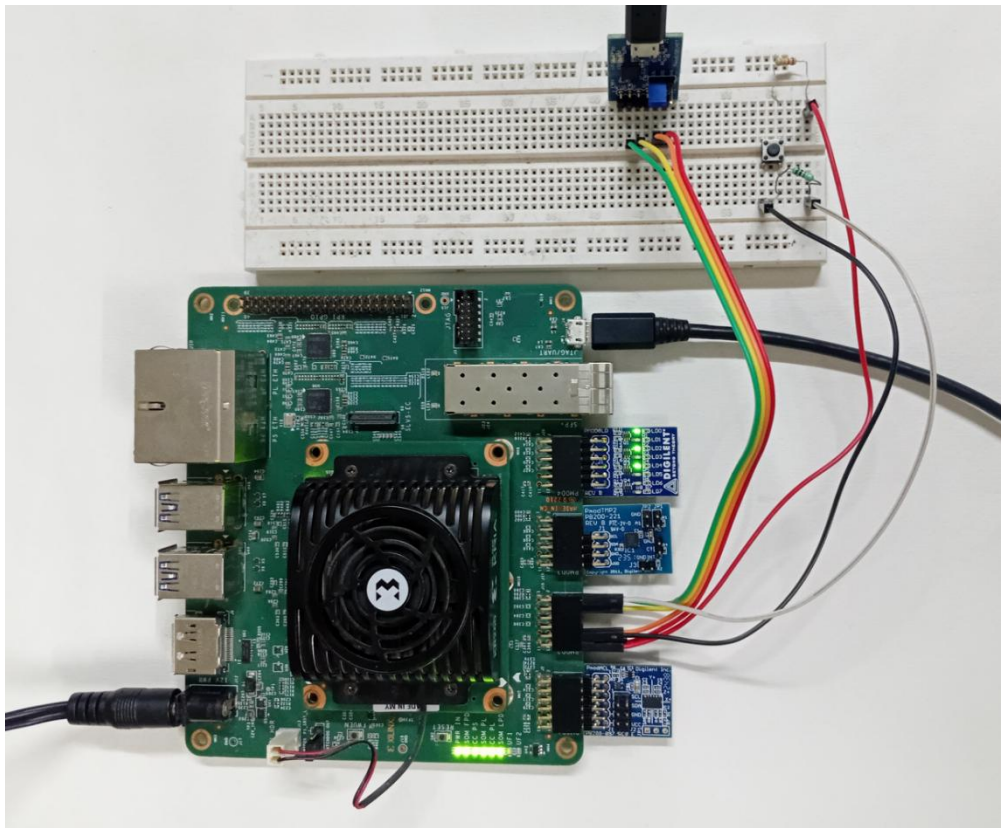
3. Connect the sensors
    a. Pmod Acl - Accelerometer
        - Connect to Pmod1.
    b. Pmod TMP2 – Temperature sensor
        - Connect to Pmod3.
    c. Pmod GPIO – 8 bit LED
        - Connect to Pmod4.
    d. Pmod UART and PL Reset
        - Connect to Pmod2.
        - You can follow following VIVADO Constraint section for connection.

```
# UARTlite
# UART_txd PMOD #2 pin 2 (connect rx of the uart device to pin 2)
set_property PACKAGE_PIN H11 [get_ports UART_txd]
set_property IOSTANDARD LVCMOS33 [get_ports UART_txd]
```

```
# UART_rxd PMOD #2 pin 4 (connect tx of the uart
device to pin 4)
set_property PACKAGE_PIN G10 [get_ports UART_rxd]
set_property IOSTANDARD LVCMOS33 [get_ports UART_rxd]

# Reset button
# PMOD #2 pin 1
set_property PACKAGE_PIN J11 [get_ports ext_reset]
set_property IOSTANDARD LVCMOS33 [get_ports
ext_reset]
```

4. Connect the Pmod-UART and KR260 UART on the board with PC. You can use Mobazterm/Teraterm in Windows or Gtkterm in Linux.
    a. Connect the baudrate of 115200 for both of the UARTs
5. Connection now looks like:



6. Now boot the KR260 board.

B. Download and Update the Kernel

For quick install of kernel in Kria-Ubuntu 22.04 LTS , please clone the github repo and run this script - https://github.com/LogicTronixInc/MicroBlaze-V-Zephyr-Kria-OpenAMP/blob/main/Kria-Ubuntu/scripts/install_updated_linux_kernel.sh

Here are the manual steps:

1. Download the updated kernel from here using wget command at KR260 (or do the download at PC and SSH copy or USB disk based copy can be done).
2. Now follow this command to update the kernel at Kria-Ubuntu
3. Reboot the board

C. Download the Firmware and Host Code

To install the firmware, please run this script - https://github.com/LogicTronixInc/MicroBlaze-V-Zephyr-Kria-OpenAMP/blob/main/Kria-Ubuntu/scripts/install_firmware.sh

Here are the manual steps:

1. Download the firmware from here – Github link
    a. Can also do the wget for the firmware file only or perform SSH copy or USB drive based copy

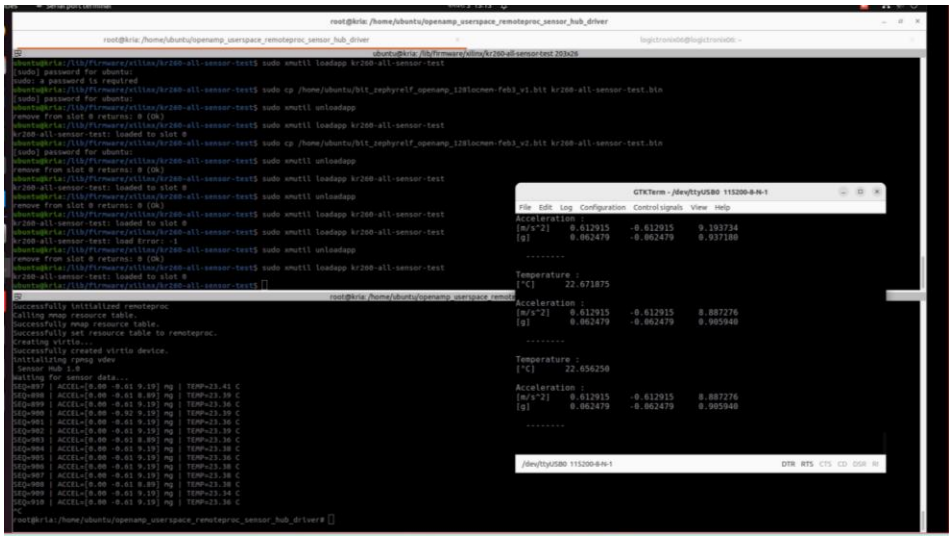
2. Copy the firmware at the /lib/firmware/Xilinx
3. Now load the firmware using xmutil command

D. Application run

Please follow this readme for running the application at KR260 Ubuntu (host app) and KR260 Zephyr (sensor fusion app) - https://github.com/LogicTronixInc/MicroBlaze-V-Zephyr-Kria-OpenAMP/tree/main/Kria-Ubuntu#running-the-application

Here are the manual steps:

1. You can connect the KR260 via SSH (for multi terminal access) as like here:



2. Now, run the OpenAMP host app on the Kria-Ubuntu terminal.
3. Now unload the kria app and re-load it.
4. Now you must able to see the output as like in above picture.

Revision History

S.N	Revision Number	Revision Details	Author
1.	v1.0	Initial Revision	Krishna Gaihre and Sanam Shakya